

Severing verbs from their arguments: Directionals and argument structure in Mam¹

Noah Elkins
Haverford College
nelkins@haverford.edu

Justin Royer
Université de Montréal
justin.royer@umontreal.ca

Tessa Scott
University of California, Berkeley
tessa_scott@berkeley.edu

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Abstract

This paper proposes an analysis of argument structure in Mam, with a focus on directional auxiliary verbs (henceforth directionals). We argue that directionals play a fundamental role for the derivation of argument structure that goes beyond their well-known contribution to spatio-temporal deixis. The evidence for this proposal comes from the observation that transitive and unaccusative verbs *require* co-occurrence with a directional, whereas unergative verbs never do. We propose a decompositional analysis of argument structure in event semantics (Davidson 1967), arguing for a neo-Davidsonian approach in which Mam verbs ought to be severed not just from their external arguments (à la Kratzer 1996), but also from their internal arguments. The result is a theory in which verbs constitute mere properties of events (as in e.g., Castañeda 1967, Parsons 1990, Levinson 2010, Elliott 2020), and so directionals are required to introduce THEMES. Finally, we show that a reinterpretation of Mam’s exceptionally rich verbal morphology provides us with overt evidence for the existence of agentive functional heads (as in Kratzer 1996), as well as evidence that the thematic domain must sometimes be split between separate vP and VoiceP projections (as in Harley 2013).

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Abbreviations in this paper follow Leipzig glosses with the following additions: A – “Set A” (ergative, possessive); AP – antipassive suffix; B – “Set B” (absolutive); COM – completive aspect; DA – disagreement suffix; DIR – directional; DS – directional suffix; MW – measure word; NF – nonfinite suffix; PAT – patient; PROX – proximate aspect; RN – relational noun; TV – transitive voice suffix; v.AGT – agentive v morpheme. In some cases glosses from other works have been adapted for consistency.

1 Introduction

Many Mayan languages employ directional auxiliary verbs, henceforth “directionals”, in the expression of verbal or stative predicates (see e.g., [England 1976, 2001](#); [Martin 1977](#); [Haviland 1993](#); [Zavala 1993](#); [Aissen 1994](#); [Mateo Toledo 2004, 2008, 2012, 2023](#); [Henderson 2016](#); [Aissen, England, and Maldonado 2017](#); [Henderson, Elias, Royer, and Coon 2018](#); [Elias 2019](#)). Among other uses discussed below, these items are best known to contribute information pertaining to the direction, location, or time of an event. A near minimal pair from the Mam dialect of Todos Santos Cuchumatán is provided below.

- (1) a. Ma Ø **pon** t-ii-’n Noé kiḥ
 PROX B2/3SG DIR:arrive.there A2/3SG-bring-DS Noé fish
 ‘Noé took the fish there.’ (TS Mam)
- b. Ma tz=**ul** t-ii-’n Noé kiḥ
 PROX B2/3SG=DIR:arrive.here A2/3SG-bring-DS Noé fish
 ‘Noé brought the fish here.’ (TS Mam)

In (1a), the directional *pon* combines with the verbal root *ii* ‘to bring’ to convey that the AGENT, *Noé*, brought the fish to a location away from the speaker. With the directional *ul* in (1b), on the other hand, it is understood that the fish was brought towards the speaker.

The main goal of this paper is to argue that directionals in Mam have functions that go beyond that of deixis, namely playing a fundamental role for argument structure. This echoes other work on Mayan languages, which have argued that directionals are needed for different reasons broadly relating to argument structure ([Mateo Toledo 2008, 2023](#), [Henderson et al. 2018](#), [Elias 2019](#)). Here, we specifically argue that directionals are needed in Mam to introduce *internal arguments*, showing that the following empirical generalizations hold in at least three dialects of Mam.

- (2) Generalizations about directionals in Mam:
- a. All (but one) transitive verbs require a directional;
 - b. All (underived) unaccusative verbs require a directional;
 - c. Unergative verbs never require a directional;
 - d. Voice derivations (antipassive and passive) circumvent the need for directionals.

Some analyses of transitive and unaccusative verbal roots in English ([Kratzer 1996](#)), or even in Mayan languages (see e.g., [Coon 2019](#), [Coon and Royer 2020](#)), have proposed that they are lexically specified to select for an internal argument. Since [Kratzer 1996](#), it is commonly assumed that internal arguments differ from external arguments in that way. Namely, internal arguments are proposed to be *severed* from the semantics of verbal roots. With this in mind, the generalizations in (2a-2c), taken together, point to a natural hypothesis about the need for directionals in Mam. If internal arguments are in fact severed from the semantics of verbal roots, then perhaps directionals instantiate the ingredient needed to introduce them, rendering Mam verbal roots essentially argumentless. This is exactly what we propose in this paper:

- (3) Main proposals about Mam argument structure:
 - a. Verbal roots denote properties of events, e.g.: $\llbracket feed \rrbracket = \lambda e.FEED(e)$;
 - b. Internal arguments are introduced by a functional head, “DIR,” headed by the directional auxiliary verb;
 - c. External arguments are introduced by a functional head, v , lexically specified as $-n$.

The paper is structured as follows. Section 2 provides basic morphosyntactic background on Mam, as well as information about data and methodology. Section 3 then proposes to disentangle two uses of directionals: “lexical uses,” describing their deictic contribution; and “functional uses,” describing their relation to internal argument introduction. Section 4 zooms in on the functional use of directionals, providing data supporting the generalizations in (2a-2c). Section 5 then turns to the components of the analysis in (3a-3b), and also discusses the generalization in (2d). As we will show, (2d) not only provides evidence for our analysis, but also leads to novel insight for the syntactic analysis of passives in Mam. Section 6 then zooms into the introduction of external arguments, component (3c) of our proposal. We argue that external arguments must also be severed from the semantics of verbal roots, and propose that the introduction of external arguments—just like that of internal arguments—is signaled by overt morphology in the language, in this case $-n$. That section will also present evidence that the thematic domain in Mam must be split between separate vP and VoiceP projections (as in Harley 2013). Section 7 concludes.

2 Basics of Mam syntax

Mam (iso 639: *mam*) is a Mamean-branch Mayan language spoken predominantly in Guatemala, but also in large diaspora communities across North America, by over 500,000 speakers (Richards and Macario 2003). The data in this paper come from several varieties, either by means of data collection by one of the authors (using a hypothesis-driven fieldwork methodology; e.g., Matthewson 2004, Davis, Gillon, and Matthewson 2014, Bochnak and Matthewson 2020) or from published sources:

- (4) Mam varieties discussed in this paper:
 - a. San Juan Atitan (SJA) Mam (author fieldnotes)
 - b. Todos Santos (TS) Mam (author fieldnotes)
 - c. San Ildefonso Ixtahuacán (Ixt) Mam (Nora England’s publications)

In broad focus declaratives, Mam’s word order is VSO (England 1991), although $\bar{A}$ -movement operations (focus, *wh*-questioning, relativization) may move elements to the clausal periphery. A typical broad focus transitive clause is given below, showing V, S, and O constituents. Note that the element labeled ‘V’ is a complex constituent that may contain a number of elements along with the verb stem, including aspect/mood marking, person and number agreement, and directionals.

- (5)

V	S	O
[Nti’ ma tz’=ok ky-ke’y-an]	[qa xjal]	[jun ja]
[NEG PROX B2/3SG=DIR:in A2/3PL-see-DS]	[PL person]	[INDF house]
‘The people didn’t see a house.’		

(SJA Mam; Scott 2023: 56)

Mam is an ergative-absolutive language, with grammatical relations established through verbal agreement. Ergative (**Set A**) cross-references transitive subjects, whereas absolutive (**Set B**) cross-references transitive objects and intransitive subjects.² In Mam, absolutive agreement precedes ergative agreement, and, when present, the directional intervenes between absolutive and ergative agreement. We present a transitive verbal template in (6a), and two exemplar sentences showing the realization of this template: in (6b), we see a transitive clause with ergative and absolutive agreement, along with a directional; in (6c), we see an intransitive clause with only absolutive agreement and no directional.

- (6) a. Transitive verb template:
 NEG/ASP – **Set B (ABS)** – DIR – **Set A (ERG)** – ROOT – SUFFIXES
- b. Ma **chin** ok **t-tzeeq'**a-n=a.
 PROX B1SG DIR A2/3SG-hit-DS=DA
 'You hit me' (Ixt Mam; England 1983a:2)
- c. Ma **chin** b'eet=a.
 PROX B1SG walk=DA
 'I walked' (Ixt Mam; England 1983a:2)

Last to note are the verbal suffixes that occupy the rightmost position within the verb stem (excluding any enclitics that are more rightward). These suffixes include voice morphology and other types of verbal derivation, as well as one of two suffixes that track transitivity. First, all transitive verbs (with just one exception being *il* 'see') surface with the glottalized nasal transitive suffix *-n* (glossed "DS" for "directional suffix" for the moment; this will be decomposed and modified in section 6); for phonological reasons, the glottal component may surface within the verb stem (see England 1983b:53-54). Second, the vast majority of unergative intransitives surface with the intransitive nasal suffix *-n* (glossed "AP" for "antipassive"; again, we modify this in section 6), whereas unaccusative intransitives bear no dedicated suffix.

3 Two contributions for directionals in Mam

In this section, we describe Mam directionals in more detail, and make explicit their two contributions. We will ultimately propose that these two contributions—the "lexical" and the "functional"—be disentangled.

Mam features a suite of 12 directionals, presented in the table below. Directionals fit into three major categories, as discussed in Elkins 2023. The first category are the egocentric directionals (e.g. *tzaj* 'come, towards speaker'), which refer to verbal motion with respect to the speaker as the deictic center. The second category are the cardinal directionals (e.g. *jaw* 'up; to the north'), which describe verbal motion without the speaker at the deictic center. The third and final category contains just the directional *b'aj* 'complete', which does not refer to direction of motion *per se*, but instead to a completed action.

²In SJA Mam, Set B does not cross-reference the features of transitive objects. Instead, Set B on transitive verbs always appears in the default 2/3SG form, and the features of the object are realizing in object position, following the subject. The details are not relevant for the present analysis, and we refer the reader to Chapter 3 of Scott (2023) for discussion and analysis of this pattern.

Table 1: Directionals in San Juan Atitán Mam; Scott 2023: 66

Directional	Meaning	Directional	Meaning
<i>xi'</i>	go, away from speaker	<i>el</i>	out; to the west
<i>tzaj</i>	come, towards speaker	<i>ok</i>	in; to the east
<i>ul</i>	arrive here	<i>kyaj</i>	remain
<i>pon</i>	arrive there	<i>aj</i>	return
<i>jaw</i>	up; to the north	<i>iky'</i>	pass
<i>kub'</i>	down; to the south	<i>b'aj</i>	complete

The set of 12 directionals given above are all derived from intransitive verbs of motion, and many have undergone phonological truncation, such as the reduction of a long vowel into a short vowel. The intransitive verbs from which the directionals are derived are still present in the language, as shown in (7). Here, the directional *ul* ‘arrive here’ is clearly derived from the intransitive verb *uul* ‘arrive here’. The verb *uul* in (7) is not functioning as a directional auxiliary, but as an intransitive (directional) verb of motion.

- (7) Ma **tz=uul** Noé.
 PROX B2/3SG=arrive.here Noé
 ‘Noé arrived here.’ (TS Mam)

Directional auxiliaries must combine with a main verb, preceding that main verb and appearing between **Set B** and **Set A** morphology. A directional is often pronounced, along with the relevant **Set B** morphology, as its own prosodic word, as in (8) below.

- (8) Ma **tz=ul** **t-ii-**’n Noé kiḥ.
 PROX B2/3SG=DIR:arrive.here A2/3SG-bring-DS Noé fish
 ‘Noé brought the fish here.’ (TS Mam)

The suite of 12 directionals given in Table 1 may additionally be stacked; that is, two or more directionals may occupy the directional slot in the verb template. When directionals are stacked in this manner (forming ‘complex’ directionals), there is a predetermined linear order of the stacked elements: a cardinal directional precedes an egocentric one, and an egocentric one precedes *b'aj* (England 1976; Scott 2023: Table 2.26). When stacked, directionals typically undergo further phonological truncation. In example (9a), the directionals *kub'* ‘down’ and *xi'* ‘go, away from speaker’ become *ku'x*; in (9b), *jaw* ‘up, to the north’ and *tzaj* ‘come, towards speaker’ become *jatz*. We represent complex directionals in our examples as stacked clitics.

- (9) a. Ma Ø **ku'=x** n-awa-’n kjo’n
 PROX B2/3SG DIR:down=DIR:in A1SG-plant-DS corn
 ‘I planted the corn’ (TS Mam)
- b. Ma Ø **ja=tz** n-baq'o-’n jun mata is
 PROX B2/3SG DIR:up=DIR:come A1SG-harvest-DS INDF MW potato
 ‘I harvested some potatoes’ (TS Mam)

3.1 The “lexical” contribution of directionals

As mentioned above, we argue in this paper that directionals have two major contributions, which we aim to disentangle and describe separately. We begin with what we refer to as their “lexical” contribution. For some verbs, directionals seem to only add “directional”, “locative” or “aspectual” content to the verbal event. For example, certain verbs like *ajqelil* ‘run’ do not require directionals at all. However, as we see in (10b-10c), the addition of a directional adds a more explicit direction of motion to the verbal event.

- (10) a. Ma chn= ajqel-n=i.
 PROX B 1 SG= run-AP=DA
 ‘I ran.’ (SJA Mam)
- b. Ma chn= **el** ajqel-n=i.
 PROX B 1 SG= DIR:out run-AP=DA
 ‘I ran out.’ (SJA Mam)
- c. Ma chn= **ok=x** ajqel-n=i.
 PROX B 1 SG= DIR:in=DIR:go run-AP=DA
 ‘I ran in.’ (SJA Mam)

It is important to flag again here that the verb ‘run’ in (10) does not require a directional; that is to say (10a) is a complete, grammatical utterance. It is also the form that is offered in context-neutral translation tasks. Additionally, when a directional is inserted, there is a transparent semantic contribution, namely direction of movement.

3.2 The “functional” contribution of directionals

We saw above that for some verbs, directionals are not required; if they are added, they have a transparent semantic or “lexical” contribution. For other verbs, though, directionals *are required*. For example, the transitive verb *k’eyil* ‘see’ may not be used unless a directional is also present within the verbal complex. The directional that typically co-occurs with this verb is *ok* ‘in’, which has a less transparent semantic contribution; compare this with the straightforward semantic contributions of the directionals in (10b-10a).

- (11) Ma tz=*(**ok**) q-ke’y-n=i a=y.
 PROX B 2/3 SG=DIR:in A 1 PL-see-DS=DA DET=DA
 ‘We saw you.’ (SJA Mam; Scott 2023: 67)

In contrast to what we have called the “lexical” contribution of directionals, the directional in (11) is required for grammaticality. When used in this way, directionals often do not have a straightforward transparent semantic contribution with regards to the motion of the verbal event. We refer to this as the “functional” contribution of directionals.

The main focus of this paper is to analyze the functional contribution of directionals; that is, what is the purpose of directionals in active sentences like (11)? In the next section, we show that certain classes of verbs can be distinguished based on their behavior with respect to directionals. Specifically, only transitive verbs and unaccusative verbs require directionals. In contrast,

unergative verbs, as in (10), do not require directionals to form grammatical clauses. Since unaccusative and transitive verbs differ from unergative verbs in involving THEME arguments, the natural hypothesis that arises given these data is that directionals in sentences like (11) are needed to introduce THEMES. The aim of section 5 will then be to account for this fact.

4 Directionals and argument structure

We discuss three predictions that follow if directionals are needed to introduce THEME arguments in Mam:

1. Transitive active verbs should require directionals → 99% true; §4.1.
2. Unaccusative active verbs should also require directionals → true; §4.2.1.
3. Unergative active verbs should never require directionals → true; §4.2.2.

Below we demonstrate that all these predictions are borne out. Section 4.1 focuses on transitive verbs, showing that they always require directionals (though we discuss one exception). Then, we turn to intransitive verbs, showing that there is an asymmetry with respect to the behavior of unaccusatives and unergatives: whereas unaccusative verbs require directionals (section 4.2.1), unergative verbs never do (section 4.2.2). Again, the empirical generalization that emerges is that directionals, at least with active verbs (we discuss other voices in section 5), are needed whenever a THEME is present.

4.1 All (but one) transitive verbs require a directional

Beginning with transitive clauses, it has long been known that directionals are required to co-occur with transitive verbs. This observation was first made by England (1976), who summarizes the state of affairs as follows.

Most transitive verbs in Mam almost always require directionals. While theoretically possible to use these verbs without a directional, the native speaker much prefers to use one, and even has difficulty accepting forms without directionals for most transitive verbs. Therefore in eliciting a finite transitive verb, the form given almost always includes a directional. If the form is requested without a directional, the native speaker invariably responds with an intransitive rather than transitive form. (p. 204)

We already presented an example of a transitive clause with its co-occurring directional in (11); two more examples are provided below. Here, we see that the transitive verb *laq'ol* ‘buy’ requires directionals, although the precise choice of directional varies depending on the context and sentence.

- (12) a. Ma tz'=(el) t-laq'o-'n Xwan tx'otx'.
 PROX B2/3SG=DIR:out A2/3SG-buy-DS Xwan land
 ‘Xwan bought land.’ (SJA Mam; Scott 2023: 76)

- b. Ma Ø ***(tzaj)** t-laq'o-'n Xwan jun xkoy.
 PROX B2/3SG DIR:come A2/3SG-buy-DS Xwan one tomato
 'Xwan bought a tomato.' (SJA Mam; Scott 2023: 76)

We stress here that while the choice of a directional might sometimes “make sense lexically” (e.g., ‘run out’ in 10b), its core lexical content is often opaque when required by the verb, as above. Table 2 provides a sample of transitive verbs; the directionals we indicate in the table are those which seem to be one of the most often used for each verb in elicitation sessions.³

Table 2: Sample of transitive verbs in Mam

Directional + Verb	Meaning	Translation
*(ku'x) + awa'n	down.go + plant	plant
*(tzaj) + laq'o'n	come + buy	buy
*(ok) + b'yo'n	in + hit	hit
*(ok) + ke'yan	in + see	see
*(tzaj) + q'i'n	come + take	bring
*(jaw) + xk'lo'xan	up + wrap	wrap
*(kub') + qesa'n	down + cut	cut
*(xi) + chmo'n	go + weave	weave
*(xi) + yek'an	go + show	show
(*DIR) + il	(*DIR) + see	see

As foreshadowed above, the only clear exception to the principle that Mam transitives require directionals is *il* ‘see’. In fact, as indicated in (13), this verb seems to *never* allow a directional (see also Scott 2023: 69). We will return to this exception below in section 5; for now, however, it is important to note that there is another verb meaning ‘see’, *ke'yl* in (14), which is not exceptional in this way. In other words, the exception is specifically with the root *il*, whose lexical idiosyncrasy happens to disallow directionals.

- (13) Ma Ø **{*xi',*tzaj,*kub'...}** w-il xuu.
 PROX B2/3SG DIR:go;DIR:come;DIR:down A1SG-see woman
 ‘I saw the woman.’ (TS Mam)

- (14) Ma tz'-***(ok)** n-ke'y-n=i a=y
 PROX B2/3SG-DIR:out A1S-see-DS=DA DET=DA
 ‘I saw you.’ (SJA Mam; Scott 2023: 69)

4.2 Intransitives: unergatives vs. unaccusatives

We now turn to the behavior of intransitive verbs in Mam with respect to their interaction with directionals. We find an asymmetry between unergative and unaccusative verbs, in which the former does not require directionals but the latter does.

³In Table 2, some of the morphology from the verbal complex has been removed, namely, aspect and both Set A and B agreement. However, the verb roots are shown with the suffix *-n* that we gloss as “DS” for now. Keeping this morphology helps to illustrate that this suffix is found only in transitive contexts.

In terms of our categorization of unergative vs. unaccusative verbs, we find that they are clearly semantically and morphologically distinct. Unergatives denote agentive events and end with a nasal suffix *-n*, which is also found on derived antipassive verbs (and as such has historically been glossed “AP” for ‘antipassive’):

- (15) Ma qo scha-**n**=i.
 PROX 1PL play-AP=DA
 ‘We played.’ (unergative; SJA Mam)

On the other hand, unaccusative verbs in Mam denote non-agentive, directed motion, or change-of-state events, and do not end with *-n*:⁴

- (16) Ma ∅ ***(kub’)** pax xk’utz’ib’il.
 PROX B2/3SG DIR:down break computer
 ‘The computer broke.’ (unaccusative; SJA Mam)

Importantly, these two classes of verbs show differential behavior with respect to directionals. Whereas unaccusatives, or at least those that are underived (we explain what we mean by ‘underived’ below), require a directional (§4.2.1), unergatives never require one (§4.2.2).

4.2.1 Unaccusative verbs require a directional

Beginning with unaccusative verbs in Mam, we find that they require directionals. Table 3 shows a representative sample of these verbs and the directionals that typically co-occur with them.

Table 3: Underived unaccusative verbs

Directional + verb	Meaning	Translation
*(<i>jaw</i>) + <i>tolj</i>	up + fall	fall (animate)
*(<i>etz</i>) + <i>tz’aq</i>	out.come + fall	fall (inanimate)
*(<i>kub’</i>) + <i>pax</i>	down + break	break
*(<i>el</i>) + <i>neq’t</i>	out + melt	melt
*(<i>txi</i>) + <i>tzaq</i>	go + rip	rip

While an example sentence with a basic unaccusative verb was already provided in (16), we provide a second one in (17).

- (17) Ma tz’=***(e=tz)** tz’aq k’ab’il t-wi’ mes.
 PROX B2/3SG=DIR:out=DIR:come fall cup A2/3S-RN:top table
 ‘The cup fell off the table.’ (unaccusative; SJA Mam)

We note that it is important to distinguish the behavior of “underived unaccusatives” (as above) from unaccusatives derived from **transitive** verbs via passive morphology. As shown in (18a), directionals are actually *optional* with unaccusative verbs like *b’uch-j* ‘shattered’, which are derived

⁴Note that the inventory of unergative vs. unaccusative verbal roots in Mam resembles those reported for other Mayan languages, including Tsotsil (Haviland 1994); Ch’ol (Gutiérrez Sánchez 2004; Coon 2010), and Chuj (Coon 2019). Moreover, similar morphological facts as in Mam obtain in some of these languages. In Chuj, for example, unergatives are derived with overt morphonology *-w* (AP), but not unaccusatives (Coon 2019).

from a passive suffix (here *-j*) (compare the active use of this root in (18b), in which the directional is obligatory). These data are treated with more depth in section 5.4.2. In short, we will suggest that directionals are no longer obligatory in such cases, because passive morphemes are also tied to the introduction of THEMES. We additionally find that without derivation, roots like *b'uch* are only possible as stative (non-verbal) predicates (18c).

- (18) Unaccusative verb derived with passive morphology
- a. Ma Ø **(kub')** b'uch-j vas.
PROX B2/3SG DIR:down shatter-PASS glass
'The glass shattered.' (passive; SJA Mam)
- b. Ma Ø ***(kub')** b'u'ch-an Pey vas.
PROX B2/3SG DIR:down shatter-DS Pey glass
'Pey shattered the glass.' (transitive; SJA Mam)
- c. B'u'ch vas.
shattered glass
'The glass is shattered.' (non-verbal predicate; SJA Mam)

Lastly, note that when used as main verbs, directional verbs themselves never seem to require *another* directional, shown below; see Table 1 for full list.

- (19) Ma Ø pon=i
PROX B2/3SG arrive.there=DA
'You arrived there.' (directional as main verb; SJA Mam)

4.2.2 Unergative verbs never require a directional

The behavior of unaccusatives contrasts with what is observed for unergative verbs, which *never* require a directional (see also (10a) and (15) above). A sample of unergative verbs is provided in Table 4. Recall that unergative verbs are universally derived with the antipassive suffix *-n*.

- (20) Ma qo xnaq'tza-n.
PROX B1PL listen-AP
'We studied.' (unergative; SJA Mam)

In translation tasks with no previous situational context provided to consultants, unergative verbs are rarely if ever offered with a directional; and if one is provided, omitting the directional always results in an acceptable sentence. We propose that this optionality falls out from the *lexical contribution* of directionals: unergatives can appear with one whenever it is relevant to state information about the motion of the event, as in (21). In such cases, the deictic contribution tends to be much more transparent.

- (21) Ma chn= **el** ajqel-n=i.
 PROX B l SG= DIR:out run-AP=DA
 ‘I ran out.’ (unergative; SJA Mam)

⁵In Table 4, some of the morphology from the verbal complex has been removed, namely, aspect and B agreement. However, the verb roots are shown with the suffix *-(a)n* that we gloss as “AP” for now. Keeping this morphology helps to illustrate that this suffix is found only in agentive contexts.

⁶For a more nuanced analysis of the meanings of the different verbs for eating and drinking, we refer the reader to [England 1980](#).

Table 4: Sample of unergative verbs⁵

Verb	Translation	Directional	Translation
<i>yon</i>	wait	<i>sch'in</i>	read/shout
<i>ajqelan</i>	run	<i>tz'ib'an</i>	write
<i>b'ixan</i>	dance	<i>schan</i>	play
<i>b'itzan</i>	sing	<i>aq'nan</i>	work
<i>yolan</i>	talk	<i>xnaq'tzan</i>	study
<i>b'in</i>	listen	<i>wan</i>	eat food
<i>tan</i>	sleep	<i>lon</i>	eat fruit
<i>na'n</i>	pray	<i>kxun</i>	eat sth. crunchy ⁶
<i>lipan</i>	fly	<i>chyon</i>	eat meat
<i>b'et(an)</i>	walk	<i>k'an</i>	drink

4.3 Summary

Table 5 summarizes the empirical observations described in this section. All transitive (with one known exception) and (underived) unaccusative verbs must co-occur with directionals. Unergatives, on the other hand, never require one.

Table 5: Directionals and argument structure

	Arguments	DIR required?
Transitive	AGENT Theme	99%
Unaccusative	Theme	yes
Unergative	AGENT	no

As also indicated in the table, the necessity of a directional correlates almost perfectly with the presence of an internal argument (at least with active verb forms). That is, while transitives and unaccusatives both take internal arguments, unergative verbs do not.

In the next section, we propose that we can make sense of these empirical generalizations theoretically if Mam directionals are needed to introduce THEMES, explaining their functional contribution. In other words, Mam verbs in general do not introduce *any* arguments: they are predicates of events (as in Castañeda 1967, Parsons 1990, Levinson 2010, Elliott 2020). Directionals are the only exception: they introduce internal arguments, and are thus needed to express basic transitive and unaccusative clauses.

5 Severing the *internal* argument from Mam verbs

This section is structured as follows. Section 5.1 first provides the theoretical background on which our analysis of Mam argument structure is built. Sections 5.2 and 5.3 then provide our analysis. Finally, section 5.4 discusses voices other than the active voice and their effects on directional syntax. As we will see, in both antipassive and passive voices, directionals are no longer obligatory, which we interpret as welcome implications of our proposal.

5.1 Assumptions about argument structure and ν P syntax

Kratzer (1996), building on previous work on argument structure (Williams 1981, Marantz 1984, Parsons 1990, Hale and Keyser 1993), influentially proposed that external arguments should be severed from the lexical semantics of verbal roots in languages like English (see also Williams 2015 and references therein). Under this analysis, instead of taking both direct objects and transitive subjects as arguments, as in other related approaches (e.g., Davidson 1967), transitive verbs only take an internal argument. A classic entry for the transitive verb ‘to feed’ in Davidsonian Event Semantics (Davidson 1967), the type of semantics we will be assuming here, is provided in (22a). Kratzer proposes (22b) instead.

- (22) a. $\llbracket \text{feed} \rrbracket = \lambda x. \lambda y. \lambda e. \text{FEED}(x)(y)(e)$ (Davidson 1967)
 b. $\llbracket \text{feed} \rrbracket = \lambda x. \lambda e. \text{FEED}(x)(e)$ (Kratzer 1996)

How do external arguments enter the derivation in a theory like (22b)? Kratzer proposes that they be delegated to a functional head within the verbal domain, which we will represent as “ ν ” here (or Voice in many works; though see section 6 below):

- (23) $\llbracket \nu \rrbracket = \lambda y. \lambda e. \text{AGENT}(e) = y$

This functional head merges with the VP, a function from events to truth values (type $\langle s, t \rangle$), via a compositional rule called “Event Identification”, a variant of Predicate Modification insofar as it gives back the conjunction of two functions (Heim and Kratzer 1998).⁷ This is to allow type $\langle e, st \rangle$ functions to combine with type $\langle s, t \rangle$ functions. Throughout this paper, we will be assuming existential closure of the event argument higher in the structure.

To illustrate how the composition of the extended VP unfolds, consider the example in (24). In step ①, the V and the THEME, in this case Fido, first compose via Function Application (Heim and Kratzer 1998) to deliver a predicate true of events of feeding Fido. Second, Event Identification in step ② allows for the VP and ν to compose, creating a type $\langle e, st \rangle$ function that can now take an AGENT argument. The DP agent, here Mittie, is then introduced via Function Application in step ③.

- (24) Mittie fed Fido
- ```

graph TD
 NP3["③ νP<st>"] --> DP1["DP<e>
Mittie"]
 NP3 --> NP2["② ν'<e,st>"]
 NP2 --> nu["ν<e,st>
AGENT"]
 NP2 --> NP1["① VP<st>"]
 NP1 --> V["V<e,st>
feed"]
 NP1 --> DP2["DP<e>
Fido"]

```

③  $\llbracket \nu P \rrbracket: \lambda e. \text{FEED}(F)(e) \ \& \ \text{AGENT} = M$

②  $\llbracket \nu' \rrbracket: \lambda x. \lambda e. \text{FEED}(F)(e) \ \& \ \text{AGENT}(e) = x$

①  $\llbracket VP \rrbracket: \lambda e. \text{FEED}(Fido)(e)$

<sup>7</sup>As discussed in Londahl 2014 and Elliott 2020, it is not in fact necessary to resort to Event Identification as a compositional tool. One could simply posit a higher type to the argument introducing head, such that  $\nu$  may take a predicate of events as its first argument, as in (i). Our analysis is compatible with both kinds of approaches.

(i)  $\llbracket \nu \rrbracket = \lambda P_{\langle s, t \rangle}. \lambda y. \lambda e. \text{AGENT}(e) = y \ \& \ P(e)$  (adapted from Elliott 2020: (31))

With this theoretical background in place, we now turn to Mam verbs. We have two aims. First, we must understand how Mam verbal roots and directional roots are lexically specified. In particular, given the hypotheses put forth in section 4, the entry for regular verbal roots in Mam should differ from the one in (22b), insofar as internal arguments should be dissociated from lexical verbs and instead incorporated into the semantics of directional roots. Second, we must provide a syntactic composition for the extended verbal domain that integrates directionals. The next two subsections address these two aims, respectively.

## 5.2 Mam verbal roots and argument structure

As we saw in section 4, the vast majority of Mam verbs (with one exception) cannot themselves introduce internal arguments. As foreshadowed above, we propose here that this is because internal arguments in Mam are also severed from verbal roots, which merely denote properties of events.<sup>8</sup> Consider, for instance, the following entries for transitive and unaccusative verbal roots, which are essentially argumentless:

- (25) a.  $\llbracket \textit{feed} \rrbracket = \lambda e. \text{FEED}(e)$  (Mam transitive verb)  
 b.  $\llbracket \textit{melt} \rrbracket = \lambda e. \text{MELT}(e)$  (Mam unaccusative verb)

Notice in (25) that transitive and unaccusative verbs have the same basic argument structure. This follows proposals such as Kratzer (1996) or Davis (1997) for languages like English or St’át’imcets, but also the proposal in Coon 2019 for the Mayan language Chuj, in which transitive and unaccusative roots are not formally distinguished in terms of their argument structure (see also Coon and Royer 2020). As discussed by Coon (2019) (citing Levin and Rappaport-Hovav 1995), we assume that transitive and unaccusative roots can be semantically distinguished with regards to whether or not they are compatible with external causation. However, the entries for Mam verbs differ from those proposed for English (by Kratzer) or Chuj (by Coon) in that they essentially have *no* argument structure: they merely denote properties of events (compare (25a) with (22b)).

We also saw in section 4 that whenever an active verb combines with an internal argument (again with one exception), a directional is needed. To account for this, we propose that the introduction of internal arguments is delegated to directional auxiliary verbs. In particular, we propose the following denotation for directionals:<sup>9</sup>

<sup>8</sup>It is important to note that other work has also proposed this, even for English. See for instance Castañeda 1967, Parsons 1990, Levinson 2010 and Elliott 2020. Here, we maintain the analysis by Kratzer (1996) for English, as it allows to make sense of the difference between English and Mam, and Mayan-internally, Mam and Chuj (see Coon 2019). Allowing variation in whether internal arguments may be inherently selected by verbs also allows us to handle Mam-internal variation, and in particular, exceptional verbs like *il* ‘see’, which never combine with a directional (see data in (34) and (35) and the paragraph above it). To be clear, however, a null functional head could also be responsible for introducing the internal argument, even in languages like English and Chuj.

<sup>9</sup>As discussed in section 3, directionals seem to have two distinct contributions in Mam: their lexical versus functional contributions. When used purely lexically, it must also be the case that directionals do not introduce THEMES. Our analysis thus points to possible ambiguity among the closed class of 12 directionals, between directionals that introduce internal arguments and those that do not. Another way to account for the purely lexical use without resorting to an ambiguity could to posit that directionals can spellout syntactic spans or chunks of structure involving DirP and an FP that introduces the theme (see e.g., Starke 2009; Svenonius 2016). We also saw that when used functionally, the semantic contribution of directionals is sometimes bleached. We hypothesize that directional auxiliary verbs might in fact be undergoing a stage of grammaticalization, in which their directional meaning sometimes gets lost (in a way reminiscent to phrasal verbs in English). We leave an exploration of these different topics for future work.

$$(26) \quad \llbracket \text{DIR} \rrbracket = \lambda x. \lambda e. \text{DIR}(e) \ \& \ \text{THEME}(e) = x \quad (\text{Mam directional})$$

Under this account, directionals are functional items in the verbal domain that serve (i) to specify information about the direction, time, or location in which an event took place (abbreviated as “DIR(*e*)” in (26) and (ii) to introduce the internal argument (“THEME(*e*)” in (26)).

Having provided lexical entries for verbal roots and directional auxiliary verbs, we turn in the next subsection to the external syntax of directionals, and how the composition of the Mam verbal domain unfolds.

### 5.3 Composing the Mam VP domain

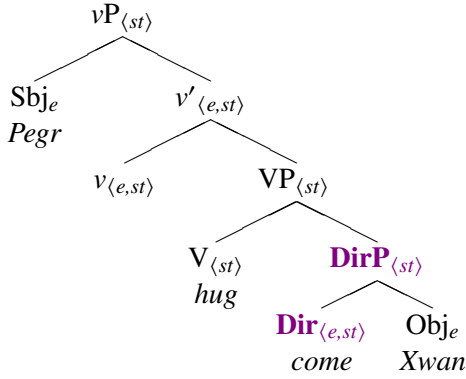
Here, we schematize how the VP domain in Mam is composed, paying special attention to the functional contribution of directionals. We provide an example of a transitive derivation, using (27) as an example.

$$(27) \quad \begin{array}{lllll} \text{Ma} & \varnothing & *(\text{tzaj}) & \text{t-xhq'e-'n} & \text{Pegr Xwan.} \\ \text{PROX B2/3SG} & & \text{DIR:come} & \text{A2/3SG-hug-DS} & \text{Pegr Xwan} \\ & & \text{'Pedro hugged Juan.'} & & \end{array} \quad (\text{TS Mam})$$

Building on Elkins (2023), we propose the minimal syntax in (28) for the extended VP domain (ignoring tense-aspect morphology). Here, the directional projection (DirP) is complement to V; it is the directional itself which selects for the internal argument of the verb.<sup>10</sup> We also make two additional assumptions regarding clause structure, which both build off previous literature on the syntax and semantics of Mayan languages. First, verb-initiality in Mam is derived via head-raising of the verb to a position at least above *v*P (following Clemens and Coon 2018 and Elkins 2023). For Mam specifically, we propose that directionals are also part of this head-movement chain, deriving the mirror DIR–VERB–VOICE morphological order. Second, we assume consistent object raising in Mam (Coon, Mateo Pedro, and Preminger 2014, Coon, Baier, and Levin 2021, Scott 2023), which we ignore in the trees for space. Object raising is independently needed to account for patterns of syntactic ergativity and binding (Coon et al. 2014, Coon et al. 2021, Royer 2023, Scott 2023), as well as split ergativity in non-finite clauses (Elkins, Royer, and Scott 2024). We crucially assume, however, that the object is interpreted in its base position for purposes of semantic interpretation.

<sup>10</sup>Another possible option for the structural position of the directional would be for it to attach as an adjunct of the VP, introducing the theme in its specifier. It would then compose with the rest of the verbal spine through Event Identification. In section 6, we provide one reason to adopt this lower structure: once head movement is assumed (following Clemens and Coon 2018, Elkins 2023), this merge order derives the correct linear order of the elements of the verbal complex

(28) Tree for (27) (*Pedro hugged Jwan*)



With this syntax in place, consider now the entries in (29), which assume the theory proposed in section 5.2.<sup>11</sup>

- (29) a.  $\llbracket xhq'e_{\text{HUG}} \rrbracket = \lambda e. \text{HUG}(e)$   
 b.  $\llbracket tzaj_{\text{COME}} \rrbracket = \lambda x. \lambda e. \text{TOWARD.SPKR}(e) \ \& \ \text{THEME}(e) = x$   
 c.  $\llbracket v \rrbracket = \lambda x. \lambda e. \text{AGENT}(e) = x$

In (29a), the verb denotes simply an event. The directional therefore has two roles: the first is to indicate information about the directionality of the event, and the second is to introduce the THEME. Finally, the entry  $v$  in (29c) will allow for the introduction of the agent via Event Identification (Kratzer 1996).

For illustration, a bottom-up derivation for the sentence in (28) is provided in (30-33).

- (30)  $\llbracket \text{DirP} \rrbracket: \lambda e. \text{TWRD.SPKR}(e) \ \& \ \text{Th}(e) = \text{Xwan}$   
  
 (through Function Application)

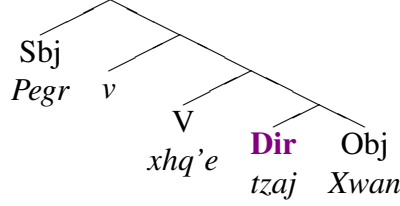
- (31)  $\llbracket \text{VP} \rrbracket: \lambda e. \text{Hug}(e) \ \& \ \text{TWRD.SPKR}(e) \ \& \ \text{TH}(e) = \text{Xwan}$   
  
 (through Predicate Modification)

- (32)  $\llbracket v' \rrbracket: \lambda y. \lambda e. \text{HUG}(e) \ \& \ \text{TWRD.SPKR}(e) \ \& \ \text{TH}(e) = \text{Xwan} \ \& \ \text{Ag} = y$   
  
 (through Event Identification)

<sup>11</sup>Note that in section 6, we propose that agentive  $v$  in (29c) is in fact overtly instantiated in Mam, via the suffix *-n*. As this section aims to focus on directionals, we have chosen to not represent this in the tree in (28). However, given the theory proposed in section 6, note that the suffix *-n* would appear under  $v$  in (28).



(33)  $\llbracket \text{vP} \rrbracket: \lambda e. \text{HUG}(e) \ \& \ \text{TWRD.SPKR}(e) \ \& \ \text{TH}(e) = \text{Xwan} \ \& \ \text{Ag} = \text{Pegr}$



(through Function Application)

Crucially, this theory of Mam argument structure proposes that internal arguments be severed from verbal roots in Mam. Whereas severing the external argument from the denotation of the verb is familiar from Kratzerian semantics, our novel contribution here is to argue that in Mam, *internal* arguments must also be severed (as also proposed in other works on English, e.g., [Parsons 1990](#), [Levinson 2010](#) and [Elliott 2020](#)).

As a final note, this analysis allows to capture the lexical exception mentioned above. Recall that the one transitive verb that does not require a directional is *il* ‘see’ (the relevant example is repeated as (34) below). We propose that this verb is unique in that it exceptionally requires a lexical entry in which the internal argument is already included (35). Being thus specified, the use of a directional is not necessary.<sup>12</sup>

(34) Ma     $\emptyset$     {*\*xi'*, *\*tzej*, *\*kub'...*}    w-il    xuu.  
       PROX B2/3SG DIR:go;DIR:come;DIR:down A1SG-see woman  
       ‘I saw the woman.’

(TS Mam)

(35)  $\llbracket il_{see} \rrbracket = \lambda x \lambda e. \text{SEE}(x)(e)$

#### 5.4 Voice and its effects on directional syntax

The theory of argument structure proposed above focused on *active* verbs. Here, we turn to the effects of voice (and valency) alternations for directional syntax. Specifically, if our proposal that directionals introduce internal arguments is on the right track, then it makes interesting predictions about the presence/optionality of directionals given different voices. In particular, we discuss two predictions, which—all else being equal—should follow from our analysis:

- (36) a. In voices in which THEMES are demoted, such as antipassives, directionals should no longer be needed. (borne out)
- b. In voices in which THEMES are present, such as passives, directionals should *a priori* still be needed. (not borne out)

In the next subsections, we show that while prediction (36a) is borne out (§5.4.1), prediction (36b) is not (§5.4.2). In other words, directionals are optional not only in the antipassive voice, as expected, but also in the passive voice. While this may seem unexpected given mainstream approaches to passives, we do not view the optionality of directionals in passives as a critical argument against our proposal. Instead, we believe that Mam’s rich morphology, and in particular its

<sup>12</sup>In fact, directionals are disallowed completely with the verb *-il*. We hypothesize that the meaning of the root *-il* itself encodes directional meaning. While this verb is often translated as “see,” one speaker comments that it means something closer to “perceive.” If this verb uniquely encodes specific directionality in the meaning of the root, it could explain why it is incompatible with directional auxiliaries.

remarkable explicitness in reflecting the introduction of arguments overtly, opens the door to a new theory of passive constructions in Mam, one in which the passive morphology specifically reflects an alternative mode of introducing THEMES.

#### 5.4.1 Antipassives circumvent the need for a directional

In Mam antipassives, the object is typically demoted to oblique status – appearing as part of a relational noun phrase – as in (37a), or simply absent, as in (37b). In such cases, a directional is never required, and such constructions in fact typically never occur without one. A representative active transitive sentence with an obligatory directional is provided in (38) for comparison. Notice that the same verbal root, *b’yo* ‘hit’, is used in (37a) and (38), suggesting again that it is truly the presence of a THEME that forces the presence of a directional.

- (37) a. N=chin    b’yo-n=i    t-i=y    ew.  
 IPFV=B1SG hit-AP=DA A2/3SG-RN:PAT=DA yesterday  
 ‘I was hitting you yesterday.’ (antipassive; SJA Mam; Scott 2023: 80)
- b. Ma    chin chmo-n-i.  
 PROX B1S weave-AP=DA  
 ‘I did some weaving.’ (antipassive; SJA Mam)
- (38) N=Ø=\*(o’k)    n-b’yo-’n=i    a=y    ew.  
 IPFV=B2/3SG=DIR:in A1SG-hit-DS=DA DET=DA yesterday  
 ‘I was hitting you yesterday.’ (transitive; SJA Mam; Scott 2023: 80)

Mam antipassives are also often employed to facilitate A-bar extraction of the transitive subject (see Coon, Mateo Pedro, and Preminger 2014 and Coon, Baier, and Levin 2020 on “Agent Focus”), as shown in the contrast in (39a). Again, directionals are typically absent from such antipassivized constructions, thereby contrasting with their active counterparts (39b).

- (39) a. Ja    xin xjaal e-Ø-b’iyoo-n    [OBL t-e    jel b’alam ]  
 DEM CLF man COM-B2/3SG-kill-AP    A2/3SG-RN CLF jaguar  
 ‘It was the man who hit the jaguar’ (antipassive; TS Mam)
- b. E    Ø    \*(kub’) t-b’iyoo-’n    xin xjaal jel b’alam  
 COM B2/3SG DIR    A2/3SG-kill-DS CLF man CLF jaguar  
 ‘The man killed the jaguar’ (transitive; TS Mam)

Importantly, these facts fall naturally from the analysis of argument structure proposed in section 5. If the obligatoriness of directionals is truly tied to internal argument introduction, as we proposed, then demoting or removing internal arguments should bleed the necessity for directionals. The above examples show that this is the case.

Finally, we note here that antipassive and unergative verbal stems take the same suffix, namely *-n*. In section 6, we will relate this suffix to AGENT introduction. Before doing that, however, we turn to a second voice construction and its effects on directional syntax: passives.

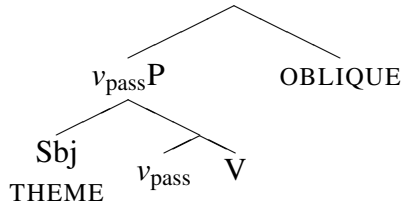
### 5.4.2 Passives also circumvent the need for a directional

Passives in Mam can be constructed from one of many derivational morphemes that appear to be in complementary distribution, including *-(e)et*, *-t*, *-j* (see [England 2017](#)). With passives, the subject is demoted and the verb bears passive morphology. This morpheme is also in complementary distribution with the suffix *-n*, used with antipassives and unergatives in the language. As for directionals, they become optional with passive morphology, and in neutral elicitation contexts, are typically provided without one. A passive-active contrast is provided below.

- (40) a. E    ∅        pj-eet    q'a Miguel [OBL t-u'n        xin xjaal ]  
           COM B2/3SG hit-PASS CLF Miguel        A2/3SG-by CLF man  
           'Miguel was hit by the man' (passive; TS Mam)
- b. E    tz'=ok        t-pju-'n        xin xjaal q'a Miguel  
           COM B2/3SG=DIR A2/3SG-hit-DS CLF man CLF Miguel  
           'The man hit Miguel' (transitive; TS Mam)

Since we proposed that directionals are needed to introduce THEMES, their optionality in passive constructions seems problematic. This is because passives always involve THEMES, in fact, such construction *promote* THEMES to subject status. In more studied languages like English, this has led to the classic treatment of passives: that they involve movement of the internal argument (or the “logical object”) to subject position ([Chomsky 1981](#), [Jaeggli 1986](#), [Collins 2005](#), a.o.).

However, once we consider together (a) the clear necessity of directionals in transitive and unaccusative constructions and (b) the evidence that roots in Mam are essentially argumentless, we believe that the Mam facts suggest an alternative analysis of passives (at least for this language), one which does not involve any movement. While we leave a full-fledged account of passives in Mam for the future, we hypothesize that passive morphology simply represents an alternative strategy for THEME introduction via passive *v*. This is schematized in (41).<sup>13</sup>

- (41) a. 
- b.  $\llbracket v_{\text{pass}} \rrbracket: \lambda x. \lambda e. \text{THEME}(e) = x$

Crucially, if passives represent an alternative means for THEME introduction, then it follows that directionals will no longer be necessary to introduce them. Again, we leave an in-depth formal analysis of Mam passives for the future.

## 5.5 Summary

This section has provided an analysis of argument structure in Mam, with a focus on the introduction of internal arguments. In particular, we argued that in the active voice, directionals are

<sup>13</sup>This proposal of course entails abandoning a strict version [Baker's](#) (1985, 1988) Uniformity of Theta Assignment Hypothesis (UTAH). Other, relative versions of UTAH could however be preserved; see e.g., [Belletti and Rizzi 1988](#)).

necessary in transitive and unaccusative clauses because only they can introduce internal arguments. In other words, internal arguments are severed from the semantics of verbal roots, and are instead delegated to directionals. We then discussed voice alternations, concluding that both antipassives and passives circumvent the need for a directional. While the former fact was predicted by our theory (since antipassives demote internal arguments), the fact that directionals are not needed with passives was unexpected. This led us to suggest a possible solution to this issue, whereby passive *v* also introduce THEMES.

In sum, we arrive at the following (tentative) inventory of argument-introducing functional heads in Mam:

- (42) a. Directionals (DIR), which introduce THEMES
- b. Passive *v*, which also introduces THEMES
- c. Agentive *v*, which introduces AGENTS

In the next and final section before the conclusion, we turn our attention to agentive *v* in (42c). Specifically, we propose that Mam is again explicit about its argument structure, requiring the morpheme *-n* whenever an AGENT is present. We thus tie this morpheme to a Kratzerian agentive *v*P.

## 6 Decomposing Mam argument structure even further

While the analysis in section 5 focused on the introduction of internal arguments, it was built on the assumption that Kratzer’s (1996) severing of external arguments holds in Mam as well. In this section, we present evidence from the morphosyntax of Mam that this is indeed true. In particular, section 6.1 argues that the morpheme *-n* is the overt manifestation of agentive *v*. Additionally, we propose in section 6.2 the *v* is split from Voice (as in Harley 2013), and that each are uniquely realized in the morphology. Finally, in section 6.3, we combine our proposals from sections 5, 6.1, and 6.2, providing our final analysis of the extended syntax of the Mam verbal domain.

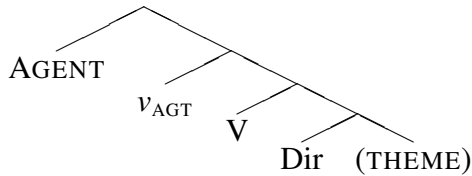
The resulting theory ultimately paints a very neat picture of how verbal derivation is structured and realized in Mam, aligning with what can be described as a recent theoretical push towards the decomposition of argument structure (Castañeda 1967, Parsons 1990, Kratzer 1996, Pytkäinen 2008, Ramchand 2008, Williams 2015, Levinson 2010, Elliott 2020, a.o.). The syntactic analysis likewise validates that our Kratzerian assumptions about external arguments are independently motivated in Mam. And while it has clear analytical power, it also requires proposing a new analysis of verbal morphemes in Mam that differs from the one assumed in decades of previous work.

### 6.1 External arguments in the verbal domain: agentive *v*

Here we argue that the morpheme *-n* is the realization of an “agentive” *v*, the head that introduces all AGENTS. As previously argued, *v* introduces AGENTS regardless of transitivity. Likewise, *-n* appears on verbs in Mam when and only when an AGENT is present in the verbal projection, regardless of transitivity. This description encompasses the following construction types: transitive

(44a), antipassive (44b), and unergative (44c).<sup>14</sup> Some of the examples in (44) are repeated from (37), but with “ $v_{\text{AGT}}$ ” in the glosses reflecting our proposal in this section.

(43) Agentive clauses in Mam (repeated from 28 and 33)



- (44) a. N=Ø=o'k                      n-b'yo-'-n=i                      a=y                      ew.  
 IPFV=B2/3SG=DIR:in A1SG-hit-TV-v.AGT=DA DET=DA yesterday  
 'I was hitting you yesterday.'                      (transitive verb; SJA Mam; Scott 2023: 80)
- b. N=chin                      b'yo-n=i                      t-i=y                      ew.  
 IPFV=B1SG hit-v.AGT=DA A2/3SG-RN:PAT=DA yesterday  
 'I was hitting you yesterday.'                      (antipassive verb; SJA Mam; Scott 2023: 80)
- c. Ma                      qo                      yol-n=i.  
 PROX B1PL speak-v.AGT=DA  
 'We talked.'                      (unergative verb; SJA Mam; Scott 2023: 266)

Passives and unaccusatives contrast with the three construction types above in that they lack AGENTS. For passives (45a), we proposed in section 5.4.2 that passive  $v$  instead introduces the sole THEME argument. For underived unaccusatives (45b), the verb must combine with a directional (or be a directional), which we proposed in section 5.2 introduces the THEME argument. Crucially, though, passives and unaccusative constructions pattern together in their morphology: both systematically lack the morpheme  $-n$ .

- (45) a. Ma                      Ø                      kub'                      qes-j                      n-q'ab'=i.  
 PROX B2/3SG DIR:down cut-PASS A2/3SG-hand=DA  
 'My hand was cut.'                      (passive; SJA Mam; Scott 2023: 83)
- b. Ma                      tz'=e=tz                      tz'aq k'ab'il t-wi'                      mes.  
 PROX B2/3SG=DIR:out=DIR:come fall                      cup                      A2/3S-RN:top table  
 'The cup fell off the table.'                      (unaccusative; SJA Mam)

From this distinction between (44) and (45), we draw the conclusion that agentive  $v$ , which introduces agents, is overtly realized as the suffix  $-n$ . In passive constructions, this head is replaced by the passive  $v$ , which has its own distinct set of morphology. If the unaccusative verb is not a directional (as in the above example), it is required to combine with a directional (DirP). If the unaccusative verb is a directional, no VP layer is required: it can introduce its sole internal argument. In either case, unaccusative clauses either lack a  $v$  projection altogether or have a  $v$  projection that is both semantically (in that it introduces no argument) and phonologically vacuous. Table 6 summarizes the forms and functions of the two flavors of  $v$  in Mam.

<sup>14</sup>A notable exception to this pattern is the verb *b'et* 'walk'. Throughout the literature, this unergative intransitive appears without  $-n$ . However, in the author's field notes on SJA Mam, there are limited but existent examples of *b'et* occurring with  $-n$ .

Table 6:  $v$  in Mam

| Head      | Morpheme      | Semantic Function | Clause types                        |
|-----------|---------------|-------------------|-------------------------------------|
| $v$ .PASS | <i>-j, -t</i> | Introduce THEMES  | Passive                             |
| $v$ .AGT  | <i>-n</i>     | Introduce AGENTS  | Transitive, Unergative, Antipassive |

One cross-linguistic consequence of this analysis is that unergatives and antipassives are derived in the same way and have the same structure. This has been argued for Kaqchikel, another Mayan language, by [Burukina and Polinsky \(to appear\)](#), who show that there is only one way to derive an intransitive clauses with one AGENT argument, encompassing both unergatives and antipassives. While this is not true for all Mayan languages (see [Coon 2019](#)), the same seems to hold in Mam.

Finally, it is important to acknowledge that segmenting *-n* as the same suffix which appears in intransitives and transitives departs from the standard analysis of Mam. [England \(1983b\)](#) analyzes *-n* in intransitives as the antipassive suffix, and glosses it “AP” accordingly. For transitives, she observes that the morpheme *-n* is always accompanied by glottalization of the preceding vowel; she thus analyzes *-’n* as a suffix that only appears on transitives. She also observes that transitives (nearly always) require directionals, and so she analyzes *-’n* as a “directional suffix” (DS). Other work on Mam, including [Scott \(2023\)](#) and [Elkins \(2023\)](#), have adopted this analysis, a convention we have also assumed in glossed examples prior to this section. Under this kind of analysis, it is a coincidence that all agentive verbal constructions share *-n*. Here, we account for this shared property by analyzing it as the realization of agentive  $v$ , which we will gloss “ $v$ .AGT” in examples of this section. In section 6.3, we turn to the glottalization only found with transitives, proposing that this feature is in fact the realization of Voice, a projection only found on transitive clauses (and therefore glossed “TV”).

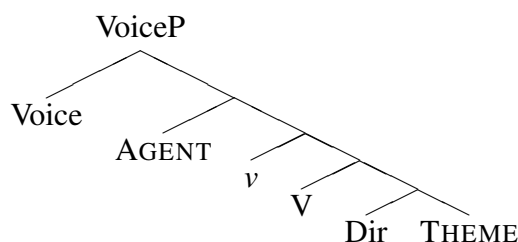
## 6.2 Voice and transitivity

In Section 6.1, we proposed that agentive  $v$  maps neatly to Mam morphology via the suffix *-n*. This now accounts for the morphology within the  $v$ P, the domain of the extended VP which we proposed in section 5 is responsible for introducing arguments. We now expand our analysis to include the Voice projection in Mam.

Before moving on, a terminological note is in order. In fleshing out the semantic composition of the verb and its arguments in section 5, the only functional verbal projection that we presented was  $v$ , which can be agentive (and introduce AGENTS) or passive (and introduce THEMES). While we call this a “voice” alternation, it is important to note that we attribute this alternation to  $v$ , and not a Voice projection. That is not to say Mam lacks Voice altogether.

In fact, in a recent analysis of a subtype of split ergativity known as “super-extended ergativity” in Mam, [Elkins et al. \(2024\)](#) independently argue that Voice, above  $v$ , is a functional layer that is only projected in transitive clauses (as in e.g., [Harley 2013](#); see also [Ranero 2021](#) and [Burukina and Polinsky to appear](#) on Kaqchikel, another Mayan language).

(46) Transitive clauses in Mam



Now, recall that Mam main verbs in transitive constructions always appear with glottalization on the final root vowel. That is, while they share with unaccusatives the requirement for directionals, and with unergatives the suffix *-n*, transitive verbs are unique in also systematically requiring glottalization. We thus propose that this glottal feature (formalized as a floating [+constricted glottis] feature) is the realization of Voice. First, we show that glottalization is unique to transitives. Then, we show that the linearization of this glottal morpheme is a phonologically conditioned, necessarily docking on the final root vowel.

The glottal feature on the final root vowel is most clearly observed in comparing transitive verbal constructions with their antipassive counterparts, since both constructions have AGENTS and thus *-n*. While the main differences in verbal morphology are that antipassives lack directionals and take Set B instead of Set A subject marking, we can also see that antipassives systematically lack final vowel glottalization.

The examples above in (44a) and (44b) of the verb *b'yool* ‘hit’ illustrate that glottalization is present with transitives but not antipassives. In addition, the two pairs of sentences below illustrate the pattern. In the transitive (a) examples below, the glottal feature, which we gloss as “TV” for “transitive”, is present. In the antipassive versions in (b), the glottal is absent.

- (47) a. N=∅=o'k                      q-b'i-'**-n**                      ...  
           IPFV=B2/3SG=DIR:in A1PL-hear-TV-v.AGT ...  
           ‘We have heard that ....’                      (transitive; SJA Mam; Scott 2023: 289)
- b. N=chin                      b'i-**n=i**.  
           IPFV=B1SG hear-v.AGT=DA  
           ‘I am listening.’                      (antipassive; SJA Mam)
- (48) a. Junt, n=∅=xi                      t-chmo-'**-n=i**                      t-i'j.  
           next IPFV=B2/3SG=DIR:go A2/3SG-weave-TV-v.AGT=DA A2/3SG-RN:PAT  
           ‘Next, you start to weave it.’                      (transitive; SJA Mam; Scott 2023: 273)
- b. N=qo                      chmo-**n=i**.  
           IPFV=B1SG weave-v.AGT=DA  
           ‘We are doing some weaving.’                      (antipassive; SJA Mam)

If Voice is only present in transitives, we expect it to be absent from passive sentences. This is indeed true in Mam: final root vowel glottalization is absent from passives. Compare the transitive imperative of *q'ama* ‘say’ in (49a) with the passive *q'ama-t* ‘is told’ in (49b).



- (49) Passives lack [+c.g.] Voice (Scott 2023: 292,290)
- a. Me'n t-xi t-q'ama-'-n=i.  
NEG.V A2/3SG-DIR:go A2/3SG-say-TV-v.AGT=DA  
“Don't say these words!” (transitive; SJA Mam)
- b. ... n=Ø=xi q'ama-t  
... IPFV=B2/3SG=DIR:go say-PASS  
‘... she is told.’ (antipassive; SJA Mam)

Similarly, if Voice is only present in transitive constructions, we expect it to be absent from fully non-finite clauses in Mam. England (2013, 2017) describes a scale of finiteness in Mam and Scott (2023) proposes the syntax for four types of semi-/non-finite clauses in Mam. One type in particular is predicted to lack Voice: “fully non-finite” clauses, which lack aspect, directionals, and agreement, and appear with a non-finite suffix *-l* instead of the agentive *v* suffix *-n*. Crucially, these clauses also lack final vowel glottalization. These fully non-finite clauses appear as complements to verbs like *xi* ‘go’ and *a* ‘start’.<sup>15</sup>

- (50) Nonfinite clauses lack [+c.g.] Voice (Scott 2023: 144)
- a. Ma chj=e'x xjal [ laq'o-l (t-e) ].  
PROX B2/3PL=go person [ buy-NF A2/3SG-RN:PAT ]  
‘The people went to buy (it).’ (nonfinite; SJA Mam)
- b. ... n=qw=a'=i [ q'olb'e-l q=ib'=i ].  
... IPFV=B1PL=start=DISAGR [ greet-NF A1PL-RN:RR=DA ]  
‘... we started to greet each other.’ (nonfinite; SJA Mam)
- c. Ma tz'=ex xjaal [ tx'ema-l si ].  
PROX B2/3SG=go person [ cut-NF firewood ]  
‘The person went to cut wood.’ (nonfinite; SJA Mam)

In sum, we arrive at the following morphological analysis of two pieces of the verbal structure in Mam, shown in Table 7. This analysis captures the pattern that the nasal suffix is present in unergatives, antipassives, and transitives. It also explains why only transitive clauses have a glottal feature.

Table 7: *v* and Voice in Mam

| Head           | Morpheme               | Semantic Function | Clause Types                        |
|----------------|------------------------|-------------------|-------------------------------------|
| <i>v</i> .PASS | <i>-j, -t</i>          | Introduce THEMES  | Passive                             |
| <i>v</i> .AGT  | <i>-n</i>              | Introduce AGENTS  | Transitive, Unergative, Antipassive |
| Voice          | [+constricted glottis] | —                 | Transitive                          |

<sup>15</sup>Nominals in fully nonfinite clauses can appear in one of two ways: (i) introduced by a relational noun, indicating oblique status, or (ii) as a pseudo-incorporated object of the verb, in which case we assume they are verbal modifiers, not verbal arguments.

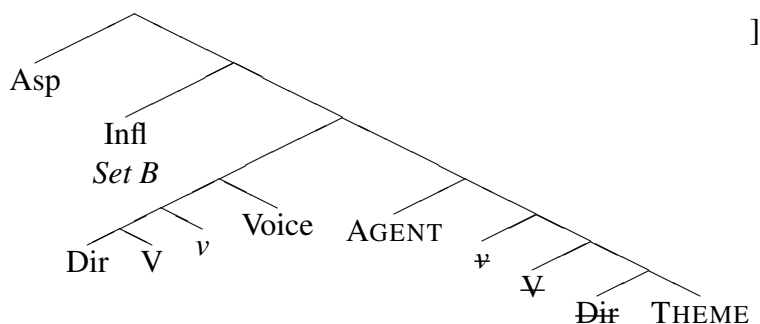
### 6.3 Putting it all together: head movement and morpheme order

So far, this section has detailed our analysis of the syntax and semantics of argument structure in Mam: we propose that directionals are required for transitives and unaccusatives because they introduce THEME arguments, which main verbs do not do. We have additionally proposed a morphological analysis of both agentive *v* and transitive Voice in the language. Finally in this section, we bring together these two analyses by sketching the syntax of the Mam extended verbal projection.

As already discussed in section 5, we adopt the head movement analysis of verbs in Mayan languages (Clemens and Coon 2018, Elkins 2023). Specifically, we adopt Clemens and Coon’s analysis that the verb head moves to a landing site above the AGENT, to a head “at the edge of the extended verbal projection that sits above the subject” and hosts morphology related to transitivity. We take this projection to be Voice, seeing no language-internal reason to diverge from Voice as the verbal projection above *v*.<sup>16</sup>

For Mam, recall the clausal spine does not begin with the VP; verbs which need internal arguments select a DirectionalP (DirP). We assume the Dir head participates in head movement through V, *v*, Voice, and lands at F. Above the F projection sit Infl and Asp, heads which Scott (2023) proposes are separate; Infl is responsible for object agreement while Asp provides aspectual information and morphology. The head movement analysis is illustrated below.

#### (51) Mam transitive syntax and morphology



An advantage of the analysis sketched in (51) is that the morpheme order that it derives is indeed the one observed for Mam, given in (52). Recall that Mam is VSO; aspect (and negation) delineate the left edge of the verbal complex, while Voice marks the right edge.<sup>17</sup> The subject and object follow the verb in that order.<sup>18</sup>

<sup>16</sup>For neutrality, Clemens and Coon label this projection “status suffix”, or “ssP,” based on the final “status suffix” morphemes indicating transitivity in Mayan languages like Ch’ol and Chuj. The authors leave out a detailed analysis of the exact nature of this projection.

<sup>17</sup>Coon (2017) argues, for the Mayan language Ch’ol, that ergative “Set A” morphemes are agreement prefixed to the Verb root. Different analyses of Set A assignment would be compatible with the structure we propose (see e.g., Deal and Royer 2023).

<sup>18</sup>Royer (2022, 2023) shows direct evidence that objects raise above subject in Mam. To derive VSO word order with high objects, we follow Scott (2023), who builds on a body of literature on VSO in Mayan in assuming rightward specifiers for the subject and the object’s landing site in Mam (see also Little 2020). This is not indicated in the tree for brevity. Additionally, we adopt Scott’s analysis of the disagreement enclitic =i as the *in-situ* partial spell out of certain pronominal arguments. When the disagreement enclitic references the subject, it appears immediately to the right of the verbal complex, correctly predicted by the present analysis.

- (52) [ Asp - Set B - Dir - Set A - V - *v* - Voice ]<sub>V</sub>                      [ AGENT ]<sub>S</sub>                      [ THEME ]<sub>O</sub>

While Voice is linearly the final head in the verbal complex, we argue that its realization is a floating [+constricted glottis (+c.g.)] feature, and thus we do not expect it to appear categorically after the other verbal morphemes. Evidence that Voice is the [+c.g.] feature and not a segmental /ʔ/ is that it is realized on the final root vowel, which may or may not be the final vowel of the verb. As shown in the examples in (53), in vowel-final verb roots, the final root vowel is glottalized.<sup>19</sup> This is true regardless of the shape/syllabic content of the root, as well as the presence and location of other glottalized consonants:

- (53) [+c.g.] Voice on V-final roots (Scott 2023: 273)

- a. Tnel, n=∅=tzaj                      q-laq'o-ʔ-n                      lan.  
first, IPFV=B2/3SG=DIR:come A1PL-buy-TV-*v*.AGT wool.thread  
'First, we buy wool thread.' (SJA Mam)
- b. Junt, n=∅=jaw                      q-q'ano-ʔ-n.  
next IPFV=B2/3SG=DIR:up A1PL-organize.thread-TV-*v*.AGT  
'Next, we organize the thread.' (SJA Mam)
- c. ... n=∅=xi                      t-kyji-ʔ-n=i                      t-txa'n.  
... IPFV=B2/3SG=DIR:go A2/3SG-twist-TV-*v*.AGT=DA A2/3SG-nose  
'... you twist the ends.' (SJA Mam)

With consonant final verb roots, the agentive *v* suffix has two allomorphs. Before the disagreement enclitic =*i*, it appears as *-n*, shown in (54a). When *-n* is word-final following a root consonant, a schwa-like vowel is epenthesized, represented orthographically as <*a*>. This epenthetic vowel does not attract the glottalization of transitive constructions. Instead, the final vowel of the root is glottalized, shown in (54b) and (54c).

- (54) [+c.g.] Voice on on C-final roots

- a. Ma    tz'=ok                      q-ke'y-**n=i**                      a=y.  
PROX B2/3SG=DIR:in A1PL-see-*v*.AGT=DA DET=DA  
'We saw you.' (SJA Mam)
- b. Ma    tz'=ok                      ky-ke'y-**an**                      qa qin=i.  
PROX B2/3SG=DIR:in A2/3PL-see-*v*.AGT PL 1SG=DA  
'They saw me.' (SJA Mam)
- c. Ma    ∅                      jaw    q-xk'lo'x-**an**.  
PROX B2/3SG DIR:up A1PL-wrap-*v*.AGT  
'We wrapped it.' (SJA Mam)

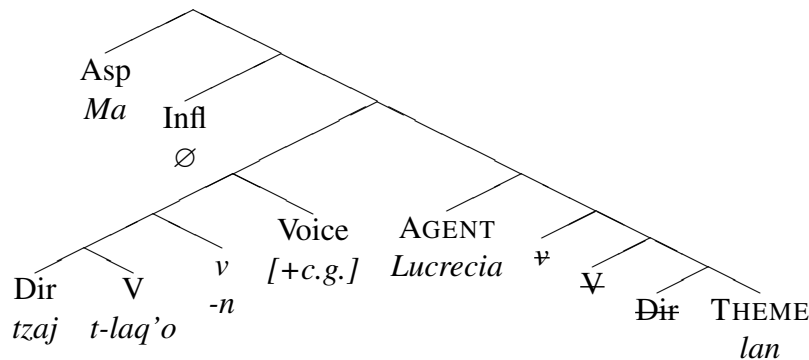
With all pieces of the analysis in place, we can now understand (i) how each argument in the Mam verbal clause is introduced, (ii) how the functional heads map to morphology, and (iii) how

<sup>19</sup>/ʔ/ does not appear to have phonemic status in Mam: it is often epenthesized to fill out a minimal CVC root requirement. The realization of glottal epenthesis differs between dialects. In Todos Santos Mam, for example, Elkins & Kuo (2022) demonstrate that in word-final position, a consonantal /ʔ/ is pronounced, and additionally undergoes a process of aspiration found on all word-final voiceless stops. After a vowel but before any additional coda consonants, /ʔ/ is realized as creaky phonation, increased phonetic duration, and a falling tonal contour on the vowel.

morpheme order is derived via head movement. A transitive clause is given in (55), and a tree illustrating our final proposal for the structure of the extended verbal domain is given in (56).

- (55) Ma     $\emptyset$         tzaj        t-laq'o'-n                    Lucrecia lan.  
 PROX B2/3SG DIR:come A2/3SG-buy-TV-v.AGT Lucrecia wool.thread  
 ‘Lucrecia bought wool thread.’ (SJA Mam)

- (56) Mam transitive clause (tree for 55)



## 7 Conclusion

In this paper, we proposed that directionals in Mam have a dedicated semantic function that goes beyond that of spatio-temporal deixis, namely that they play a fundamental role for argument structure. We showed that there is reason to believe that the denotation of Mam verbs ought to be severed not just from their external arguments (à la Kratzer 1996), but also from their *internal* arguments; with verbs constituting properties of events, directionals are required to introduce THEMES. This proposal makes it possible to account for the observation that transitives and unaccusatives require directionals, whereas unergatives do not require them.

We then showed how this analysis can be brought to bear on voice constructions such as the passive and antipassive, in which directionals are not required. Whereas the behavior of antipassives (in which internal arguments are demoted) follows straightforwardly from our approach, the behavior of passives requires some care. We adopt a novel theory of passives for Mam in which the subject is not the internal argument, but is instead a thematic subject introduced by a passive *v*.

Turning from the implications of directionals as introducers of THEMES, we presented evidence that the introduction of AGENTS may also be discovered from the verbal morphology of Mam. In particular, we can split the thematic domain into agentive *v*P and VoiceP, the latter of which is only present in transitives. First, agentive *v* is realized by the suffix *-n*, and is therefore present on unergative verbs and in antipassives. Second, Voice is realized by a floating [+constricted glottis] (abbreviated [+c.g.] throughout) feature which docks on the final root vowel.

In conclusion, we find that identifying a dedicated function for directionals for argument structure in Mam opens up a number of avenues for investigating its thematic domain more generally. Seeing as directionals in other Mayan languages have also recently been implicated in the derivation of certain verbal constructions (e.g., indirect arguments (Mateo Toledo 2004, 2008, 2023) and stative predication (Henderson et al. 2018, Elias 2019) in Q'anjob'alan languages), we hope that this work ushers in further fruitful research on directionals across the Mayan language family.

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